

The `caption` package

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Abstract

The `caption` package provides many ways to customise the captions in floating environments such `figure` and `table`. The `\rotcaption` command and the `sidewaysfigure` and `sidewaystable` environments provided by the `rotating` package from S. Rahtz and L. Barroca are also supported. The `caption` package also cooperates with the `float` package written by A. Lingnau and the `subfigure` package written by S.D. Cochran.

1 The user interface

To use this package just type

```
\usepackage[options]{caption}
```

in the preamble of your document, where the following options are supported:

normal provides ‘normal’ captions, this is the default

hang or **isu** provides captions with hanging indention

center provides captions where each line is centered

centerlast provides captions where the last line of the paragraph is centered

scriptsize, **footnotesize**, **small**, **normalsize**, **large** or **Large** sets the default size of the caption

up, **it**, **sl**, **sc**, **md**, **bf**, **rm**, **sf** or **tt** sets the default style of the caption label

ruled supports ruled floats of the `float` package, see section 1.1 for details

To setup the size and style of the captions, this package also provides the commands `\captionsize` and `\captionlabelfont` to allow a more flexible way to customize the captions than the above options could do. `\captionsize` is called before each caption, `\captionlabelfont` is called just before the label of the caption. Note that these commands are used by the options, e.g. the options **small** and **sf** are identical to

```
\renewcommand{\captionsize}{\small} and
\renewcommand{\captionlabelfont}{\sffamily}.
```

There is also a new length `\captionmargin` to setup an extra left and right margin for the captions, e.g. the command

```
\setlength{\captionmargin}{10pt}
```

sets this margin to 10pt.

1.1 The rotating, float and subfigure package

If you want to use this package together with the `rotating`[1], `float`[2] and/or the `subfigure` package, you have to input this package *after* the other ones, like

```
\usepackage{rotating,subfigure}
\usepackage[center,small,sc]{caption}
```

The `caption` package now redefines the `sidewaysfigure` and `sidewaystable` environments and the `\rotcaption` command provided by the `rotating` package from S. Rahtz and L. Barroca. Note that the `\captionmargin` does not affect the `\rotcaption` command.

It also redefines the captions of the `plain` and `boxed` styled floats provided by the `float` package from A. Lingnau. `ruled` floats are not supported by default, but you can change this via setting the option `ruled`. Note that only the `\captionsize` and `\captionlabelfont` take effect in ruled floats.

The `subfigure` package from S.D. Cochran is only supported via the options `scriptsize`, ..., `large`, in this case this package redefines the `\subcaptionsize` in an adequate way. If you redefine `\captionsize` by yourself and use the `subfigure` package, you also have to redefine `\subcaptionsize` by yourself.

This package cooperates with the version 2.0 (1994/04/15) of the `rotating` package, version 1.2 (1994/06/20) of the `float` package and version 1.6 (1993/05/13) of the `subfigure` package, but will hopefully work with future versions, too.

2 Example

Here's an example which was produced by the following code in the preamble of this document:

```
\usepackage[hang,small,sf]{caption}
\setlength{\captionmargin}{20pt}
```

EXAMPLE FIGURE

Figure 1: An example figure with a small caption and a sans serif label. The `hang` option was used. There is a margin of 20pt.

3 The code

3.1 Identification

```
1 \NeedsTeXFormat{LaTeX2e}[1994/06/01]
2 \ProvidesPackage{caption}[1994/11/28 v1.2 caption package (AS)]
3 \typeout{Package: caption v1.2 <1994/11/28> (Axel Sommerfeldt)}
```

3.2 Initial code

```
\captionsize
4 \newcommand{\captionsize}{}%
```

```
\captionlabelfont
5 \newcommand{\captionlabelfont}{}%
```

```
\captionmargin
6 \newlength{\captionmargin}
```

Here are the different basic types of captions implemented:

```
\@asnormalcaption the normal caption
7 \newcommand{\@asnormalcaption}[2]{%
8   #1: #2}
```

```
\@asisucaption the iso- or hangcaption; this code was taken from The LATEX Companion[3,
p155] and modified
9 \newcommand{\@asisucaption}[2]{%
10   \sbox{\@ascaptionbox}{#1: \ }%
11   \addtolength{\@ascaptionwidth}{-\wd\@ascaptionbox}%
12   \usebox{\@ascaptionbox}\parbox[t]{\@ascaptionwidth}{#2}}
```

```
\@ascentercaption the center caption
13 \newcommand{\@ascentercaption}[2]{%
14   \parbox[t]{\@ascaptionwidth}{\begin{center}#1: #2\end{center}}}
```

```
\@asannecaption the centerlast caption; the idea how to do this was taken from Brueggemann-
Klein[4], it is also mentioned in Kopka[5, p227]
15 \newcommand{\@asannecaption}[2]{%
16   \addtolength{\leftskip}{0pt plus 1fil}%
17   \addtolength{\rightskip}{0pt plus -1fil}%
18   \setlength{\parfillskip}{0pt plus 2fil}%
19   #1: #2}
```

3.3 Detection of the subfigure package

`\caption@subcapsize` If the subfigure package is loaded, a little message will be typeout and `\caption@subcapsize` – which sets the size of the subcaptions – will be defined.

```

20 \ifx\thesubfigure\undefined
21   \newcommand{\caption@subcapsize}[1]{}
22 \else%
23   \typeout{\space\space\space\space\space\space\space\space\space\space
24             'subfigure' package detected}
25   \newcommand{\caption@subcapsize}[1]{\renewcommand{\subcapsize}{#1}}
26 \fi

```

3.4 Declaration of options

There are four different types of captions supported: `normal`, `isu`, `center` and `centerlast`. `hang` is exactly the same as `isu`.

```

27 \DeclareOption{normal}{\let\@ascaption\@asnormalcaption}
28 \DeclareOption{isu}{\let\@ascaption\@asisucaption}
29 \DeclareOption{hang}{\ExecuteOptions{isu}}
30 \DeclareOption{center}{\let\@ascaption\@ascentercaption}
31 \DeclareOption{anne}{\let\@ascaption\@asannecaption}
32 \DeclareOption{centerlast}{\ExecuteOptions{anne}}

```

There are options for six different sizes available, they also redefine the `\subcapsize` provided by the subfigure package (if detected).

```

33 \DeclareOption{scriptsize}{%
34   \renewcommand{\captionsize}{\scriptsize}
35   \caption@subcapsize{\scriptsize}}
36 \DeclareOption{footnotesize}{%
37   \renewcommand{\captionsize}{\footnotesize}
38   \caption@subcapsize{\scriptsize}}
39 \DeclareOption{small}{%
40   \renewcommand{\captionsize}{\small}
41   \caption@subcapsize{\footnotesize}}
42 \DeclareOption{normalsize}{%
43   \renewcommand{\captionsize}{\normalsize}
44   \caption@subcapsize{\footnotesize}}
45 \DeclareOption{large}{%
46   \renewcommand{\captionsize}{\large}
47   \caption@subcapsize{\normalsize}}
48 \DeclareOption{Large}{%
49   \renewcommand{\captionsize}{\Large}
50   \caption@subcapsize{\large}}

```

There are nine options available to set the style of the label.

```

51 \DeclareOption{up}{\renewcommand{\captionlabelfont}{\upshape}}
52 \DeclareOption{it}{\renewcommand{\captionlabelfont}{\itshape}}
53 \DeclareOption{sl}{\renewcommand{\captionlabelfont}{\slshape}}
54 \DeclareOption{sc}{\renewcommand{\captionlabelfont}{\scshape}}

```



```

87   \@makeascaption{#1}{#2}}
88   \ifx\caption@ruled\undefined
89   \else
90     \renewcommand\floatc@ruled[2]{%
91       \captionsize%
92       \@asnormalcaption{{\captionlabelfont #1}{#2}\par}
93   \fi
94 \fi

```

3.8 Support of the rotating package

If the rotating package is loaded, the commands `\@makerotcaption` (for support of `\rotcaption`) and `\@makercaption`, furthermore the environments `sidewaysfigure` and `sidewaystable` will be redefined here.

```

95 \ifx\@makercaption\undefined
96 \else
97   \typeout{\space\space\space\space\space\space\space\space\space\space
98     'rotating' package detected}
99   \renewcommand{\@makercaption}[2]{%
100     \vskip 10\p@%
101     \setlength{\@ascaptionwidth}{\textheight}%
102     \@makeascaption{#1}{#2}}
103   \renewcommand{\@makerotcaption}[2]{%
104     \captionsize%
105     \sbox{\@ascaptionbox}{\captionlabelfont #1:} #2}%
106     \ifdim \wd\@ascaptionbox > .8\vsiz
107       \rotatebox{90}{%
108         \setlength{\@ascaptionwidth}{.8\textheight}%
109         \begin{minipage}{\@ascaptionwidth}%
110           \@ascaption{\captionlabelfont #1}{#2}%
111         \end{minipage}}\par
112     \else%
113       \rotatebox{90}{\usebox{\@ascaptionbox}}%
114     \fi
115     \hspace{12pt}}
116   \renewcommand{\sidewaysfigure}{%
117     \let\@makecaption\@makercaption\@rotfloat{figure}}
118   \renewcommand{\sidewaystable}{%
119     \let\@makecaption\@makercaption\@rotfloat{table}}
120 \fi

```

References

- [1] Sebastian Rahtz and Leonor Barroca: *A style option for rotated objects in L^AT_EX*, 1994/06/07
- [2] Anselm Lingnau: *An Improved Environment for Floats*, 1994/06/20

- [3] Michel Goossens, Frank Mittelbach and Alexander Samarin: *The L^AT_EX Companion*, Addison-Wesley, Reading, Massachusetts, 1994.
- [4] Anne Brueggemann-Klein: *Einführung in die Dokumentverarbeitung*, B.G. Teubner, Stuttgart, 1989
- [5] Helmut Kopka: *L^AT_EX– Erweiterungsmöglichkeiten*, 3. überarbeitete Auflage, Addison-Wesley, Bonn, 1991